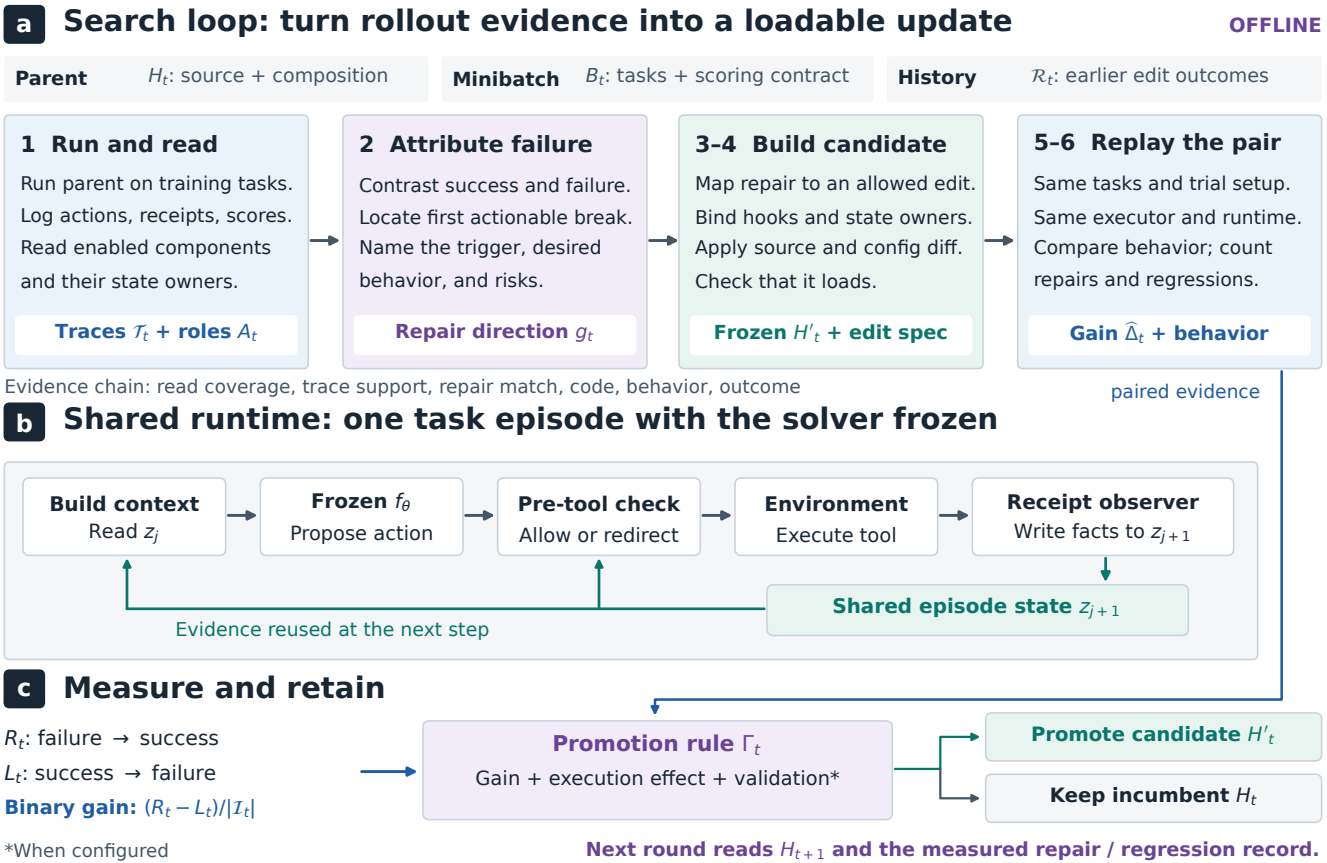

a
Search loop: turn rollout evidence into a loadable update
OFFLINE
Parent
H_t: source + composition
Minibatch
B_t: tasks + scoring contract
History
$\mathcal{R}_t$: earlier edit outcomes
1 Run and read
Run parent on training tasks.
Log actions, receipts, scores.
Read enabled components
and their state owners.
Traces $\mathcal{T}_t$ + roles A_t
2 Attribute failure
Contrast success and failure.
Locate first actionable break.
Name the trigger, desired
behavior, and risks.
Repair direction g_t
3-4 Build candidate
Map repair to an allowed edit.
Bind hooks and state owners.
Apply source and config diff.
Check that it loads.
Frozen H'_t + edit spec
5-6 Replay the pair
Same tasks and trial setup.
Same executor and runtime.
Compare behavior; count
repairs and regressions.
Gain $\widehat{\Delta}_t$ + behavior
Evidence chain: read coverage, trace support, repair match, code, behavior, outcome
paired evidence
b
Shared runtime: one task episode with the solver frozen
Build context
Read z_j
Frozen f_θ
Propose action
Pre-tool check
Allow or redirect
Environment
Execute tool
Receipt observer
Write facts to z_{j+1}
Shared episode state z_{j+1}
Evidence reused at the next step
c
Measure and retain
R_t: failure → success
L_t: success → failure
Binary gain: $(R_t - L_t)/|\mathcal{I}_t|$
Promotion rule Γ_t
Gain + execution effect + validation*
Promote candidate H'_t
Keep incumbent H_t
*When configured
Next round reads H_{t+1} and the measured repair / regression record.